Xiao Chen

Ph.D. Student in Computer Science, Indiana University Bloomington

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Research interests: Programming languages; gradual and migratory typing; operational semantics; algebraic effect handlers; formal verification.

Seeking fully funded Ph.D. opportunities outside the United States for 2027 entry.

Education

Indiana University Bloomington Sep. 2024 – Present
Ph.D. in Computer Science | Advisor: Prof. Sam Tobin-Hochstadt
Currently on leave due to prolonged U.S. visa processing delays.

Northwestern University Sep. 2022 – Jun. 2024
M.S. in Computer Science | GPA: 3.957/4.00
Research with Profs. Christos Dimoulas and Simone Campanoni.

Nanjing University Sep. 2020 – Jun. 2022
M.Eng. in Electronic Information | GPA: 4.58/5.00

Nanjing University Sep. 2016 – Jun. 2020
B.Eng. in Software Engineering | GPA: 4.28/5.00

Ongoing Research Projects

Type as Obligations (TAO)
Mechanized proofs in Rocq complete; manuscript in preparation.

- Developed an executable operational framework that separates the type obligations accumulated during execution from the checks performed by a runtime enforcement strategy.
- Mechanically derived a profiler semantics that tracks typing histories and boundary provenance through higher-order interactions, without manually instrumenting individual programs.
- Proved a bidirectional correspondence between ordinary and profiler evaluation: well-typed, uniquely labeled programs terminate with the same resolved values, despite extra bookkeeping and different location allocations.
- Defined obligation soundness and proved that it implies type soundness, but not conversely. Used violated obligations as a common reference for blame soundness and completeness.
- Compared Natural, Threesomes, Forgetful, and Transient using uniform criteria, with mechanized proofs and counterexamples in **Rocq/Coq** and an executable **Racket/Redex** model.

Migratory Typing for Algebraic Effects and Handlers
Effect-Typed-Racket / Prototype implemented; formal semantics and proofs in progress.

- Extended **Typed Racket** with algebraic effect handlers and effect typing, carrying effect information through higher-order calls, polymorphic instantiation, and module imports.
- Implemented runtime contracts for escaping-effect specifications across typed/untyped boundaries. Distinguished forwarded requests and resumed computations from effects of the handler implementation.
- Supported effect identity across modules and explicit effect-row instantiation. Current contracts check concrete effect allowances; relational effect-row parametricity remains ongoing work.
- Developing a high-level calculus and a lower-level model based on delimited continuations and contracts, aiming to establish type and effect soundness and a behavioral correspondence between them.

Honors and Selected Coursework

National Scholarship for Postgraduates, Nanjing University (top 1%, Oct. 2021).
First-class People's Scholarship (top 5%, Nov. 2019); **Third-class People's Scholarship** (top 20%, Nov. 2017 and Nov. 2018), Nanjing University.

Work Experience

Tencent Holdings Ltd. Jul. 2021 – Jan. 2022
iOS Development Intern, Technology Engineering Group | Shenzhen, China

- Developed an iOS multimedia messaging app using 5G connectivity; designed messaging card styles and interaction models, and refactored the user guide and network error detection logic.

Microsoft Software Technology Center Asia Jul. 2019 – Oct. 2019
Software Engineering Intern, Speech Recognition Group | Suzhou, China

- Automated speech recognition model training for conference scenarios using Microsoft AEther. Applied Gaussian blur and SG filtering to improve recognition accuracy in short conversations.

Skills and Languages

Programming and formal methods: Racket, Rocq/Coq, Redex, Agda, C++, Haskell, Python, Swift, LLVM IR.
Tools: Linux shell, Vim, LaTeX, Git, HTCondor.
Languages: Chinese (native), English (proficient), Japanese (JLPT N1), French (beginner).